21 The length of a rectangle is the same as the length of each side of a square.

The length of the rectangle is 4 cm more than 3 times the width of the rectangle.

The area of the square is 66 cm^2 more than the area of the rectangle.

Find the length and the width of the rectangle.
You must show all your working.

Let length = x and width = (x – 4)/3 cm.

Square area = rectangle area + 66:

1 mark
Relate the two areas

$x^2 = x(x - 4)/3 + 66$

1 mark
One variable

$3x^2 = x^2 - 4x + 198$

$2x^2 + 4x - 198 = 0$

$x^2 + 2x - 99 = 0$

1 mark
Quadratic equation

×	x	11
x	x^2	$11x$
-9	$-9x$	-99

$(x - 9)(x + 11) = 0$

1 mark
Factorise and solve

$x = 9$ or $x = -11$.

Reject –11: a length must be positive.

1 mark
Select positive root

Width = (9 – 4)/3 = 5/3 cm.

Length 9 cm; width 5/3 cm.

1 mark
9cm and 5/3cm

Check: $9^2 - 9 \times (5/3) = 81 - 15 = 66\ cm^2$.

(Total for Question 21 is 6 marks)